Screeps Terminal Messenger

Integration & Protocol Guide for Allies — Automated Cross-Colony Communication

The **Terminal Messenger** (`terminalMessenger.js`) is a lightweight, zero-dependency Screeps communication module. It allows allied players to exchange automated, engine-authenticated text messages across shards by piggybacking on `StructureTerminal.send()` descriptions with 1 energy.

Core Benefits • **Zero Setup:** Standalone file — no external packages or complex frameworks required.
• **Instant Notifications:** Incoming messages trigger in-game email notifications via `Game.notify()`.
• **Anti-Spoofing:** Sender usernames are cryptographically verified by the Screeps engine.

1. Quick Integration into main.js (30 Seconds)

1. Save `terminalMessenger.js` in your Screeps script root directory.
2. Add `require('./terminalMessenger').run();` to your main loop:

```
// main.js
const messenger = require('./terminalMessenger');

module.exports.loop = function() {
    // 1. Process incoming terminal messages & dispatch email alerts
    messenger.run();

    // 2. Rest of your colony logic (spawns, creeps, market, defense...)
};
```

2. Sending Messages in the Console

The module automatically registers `message` as a global console function at startup:

A. Basic Message Send

To send a message, call `message('PlayerName', 'Your text')`:

```
> message('vlc6wsw0l', 'Defense needed at W1N1!')
Message sent to vlc6wsw0l at E2N46 from E2N48 (freight: 12 energy):
"Defense needed at W1N1!"
```

- Looks up the player in the built-in directory.
- Finds your closest terminal with available energy (cooldown 0, unused this tick).
- Sends **1 energy** with the description formatted as `MSG:<text>`.

B. Message Length Limit (Max 96 Characters)

Screeps descriptions are capped at 100 characters. Since the protocol uses a 4-character prefix (`MSG:`), your message text can be up to **96 characters**. Messages longer than 96 characters are safely rejected before spending any energy:

```
> message('Netloc', 'A'.repeat(97))
Message rejected: length is 97 characters (maximum 96 characters, 'MSG:' takes 4).
```

C. Inspecting Contact Directory & Custom Rooms

Calling `message('PlayerName')` without text displays their registered room:

```
> message('dustinc555')
Target: dustinc555 -> E9N43. Usage: message('dustinc555', 'Your message here', [optionalTargetRoom])
```

To override or target a specific room directly: `message('Ricardo306', 'Resources sent', 'E12N51')`

3. Receiving Messages & Alerts

When an ally sends a message to any of your terminals, `messenger.run()` automatically:

1. **Sends an Immediate Email Notification:** Calls `Game.notify("[Terminal Message] from ...", 0)`, triggering an instant email alert from Screeps.
2. **Prints a Highlighted Console Banner:** Outputs a high-visibility box in your console:

```
======================================================================
✉ [INCOMING TERMINAL MESSAGE]
  From:    dustinc555 (E9N43 -> E2N46 @ tick 83755700)
  Message: "Energy transfer received. Thank you!"
======================================================================
```

3. **Maintains Rolling History:** Stores the last 50 messages in `Memory.terminalMessenger.messages`.

4. Allied Network Contact Directory

The client comes pre-configured with verified RCL 8 terminal rooms for allied players:

Player	Contact Room	RCL	Sector / Cluster
vIc6wsw0l	E2N46	RCL 8	Primary Hub
TimelessTacticianEon	E3N42	RCL 8	Also owns E3N43
dustinc555	E9N43	RCL 8	Also owns E9N44, E9N45
PhilipSchlump	E2N47	RCL 8	Neighbor to E2N46
Netloc	E2N48	RCL 8	Neighbor to E2N47 / E2N46
fR1dj	W9N46	RCL 8	Western sector
Ricardo306	E11N51	RCL 8	Eastern sector (also E12N51, E11N54)
Broden1616	E4N51	RCL 8	Northern sector (also E7N51, W7N52)
Arkemenes	W7N49	RCL 8	Western sector
SunDanceKid	E7N48	RCL 5	Pre-terminal level
Atlas_Dominion	*custom*	—	Configurable via `setAllyRoom()`

5. Managing Custom Contacts & Diagnostics

- **View Directory & History:** Run `messageStatus()` in console to view the full directory and recent incoming/outgoing messages.
- **Set or Update Ally Room:** Run `setAllyRoom('PlayerName', 'RoomName')` to add a new contact or redirect an existing one. Overrides are persisted in `Memory` across bot restarts.

6. Protocol Specification & Architecture

- **Transport:** Standard `StructureTerminal.send(RESOURCE_ENERGY, 1, targetRoom, "MSG:" + text)`.
- **Freight Calculation:** Energy freight is charged according to Screeps linear distance: `ceil(amount × (1 - exp(-distance / 30)))`. For 1 energy, freight is typically 1 energy across nearby rooms.
- **Deduplication:** The receiver tracks the latest tick watermark and boundary transaction IDs, guaranteeing that every message is alerted exactly once.